

NUST College of Electrical and Mechanical Engineering

Course: Signals and Systems Lab

Degree: 44-CE

Semester: Fall 2024

Project Title: "Audio Classification"

Project Overview:

This project involves developing a system to differentiate between various sounds using signal processing techniques. Students will implement algorithms to classify audio signals into different categories (e.g., speech, music, noise) using neural networks. This project provides foundational experience in signal processing, including filtering, pitch detection, and time and frequency domain analysis.

Project Objectives:

1. **Understand Sound Signals:** Learn about ECG signals and their significance in monitoring heart activity, focusing on identifying R-peaks within the QRS complex.
2. **Signal Pre-processing:** Apply filtering and data augmentation techniques to classify the signal into voice, music and noise through pitch detection and information.
3. **Pitch Detection:** Implement algorithms to accurately detect and shift pitch.
4. **Audio Classification:** Classify the sounds into three broad categories as 1. Speech, 2. Music and 3. Noise.

Project Components:

1. **Research and Literature Review:** Study the structure of audio signals, focusing on the pitch determination and mel-frequency co-efficients.
2. **Data Collection and Preprocessing:** Obtain sound signal data from publicly available datasets such as the Urban Sound 8k dataset. Preprocess the data by using digital filters like low-pass, high-pass, or band-pass filters. Zero-crossing rate, root mean square value and energy of the signal tone.
3. **Pitch Detection Algorithm:** Implement pitch and mel-frequency component detection algorithm in a software environment like MatLab/Python. Experiment with different detection methods to compare their accuracy and robustness in classifying various sounds.
4. **Audio Classification:** Classify the sounds into three broad categories as 1. Speech, 2. Music and 3. Noise.
5. **Testing and Evaluation:** Test the system's performance under different noise conditions and with varying noise models (e.g., Rayleigh, Nakagami-m).

6. Documentation and Presentation: Prepare a detailed report documenting the research, design, implementation, and results of the project. Demonstrate its performance with sample sound signals.

Tools and Resources:

- **Software:** Python (with libraries like NumPy, SciPy) for signal processing.
- **Datasets:** Urban Sound 8k Dataset. <https://www.kaggle.com/datasets/chrisfilo/urbansound8k>
- **Reference Materials:** Textbooks on signal processing, speech processing and online tutorials on digital filtering and pitch detection.

Learning Outcomes:

By completing this project, students will:

- Develop a practical understanding of audio signals and their use in communication systems.
- Learn to apply signal processing techniques, such as filtering and peak detection, to real-world data.
- Gain experience in algorithm development and testing for signal analysis tasks.

This project provides a focused and manageable introduction to audio classification, reinforcing key concepts in signals and systems while offering practical experience with a real-world application.

Submission Date: 10th Jan-2025

Students per group: 3 to 4

Submit a report in hard format along with soft copy and source files zipped and emailed to your instructor.

Distribution of Marks:

Accomplishment of Task	= 50
Report	= 20
Individual Viva	= 30
Total Marks	= 100

Penalty in case of Late Submission:

14th-Jan-2025 (Project will be graded out of 70%)

16th-Jan-2025 (Project will be graded out of 50%)

No Submissions acceptable during and after Final Exam week.